Parental Aspirations and Children's Education:

Econometric Evidence from Indonesia

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The following tools and aids were used in the preparation of this Master's thesis:

- LaTeX zur Dokumentenerstellung.
- R für Datenbereinigung, -analyse und ökonometrische Schätzung.
- ChatGPT (OpenAI) Unterstützung beim Debuggen von Code, Überprüfung von Grammatik und Syntax sowie Hilfe bei der LaTeX-Formatierung

Abstract

This thesis studies how parental aspirations influence child education investment. The theoretical model derived hypothesizes an inverted U shape where low aspirations and extremely high aspirations discourage investment while moderate aspirations encourage investment. Using data from the Indonesian Family Life Survey (5th Round), this hypothesis is tested yielding strong evidence in favour. Used here is an original method of quantifying aspirations this work contributes to literature on aspirations and reference dependent preferences. ‘

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1 Introduction

Household decisions about investing in children's education are central to both individual well-being and aggregate development. Anecdotally, education is often viewed as a necessary investment toward later life satisfaction. Evidence shows that additional schooling and related investments translate into higher earnings and improved life chances later in life. School construction in Indonesia, for example, yielded sizeable wage returns for affected cohorts (Duflo 2001). Parental investments are an important determinant of future cognitive and socio-emotional skills of children (Attanasio et al. 2020). Access to additional coaching, tutoring, group activities, and other investments have beneficial long run impacts and were found to significantly increase annual earnings in adulthood(Lavecchia, Oreopoulos, and Brown 2020). Beyond benefits to the individual, education investments also have strong societal benefits. Increased schooling subsidies reduces adolescent pregnancies and marriage (Duflo, Dupas, and Kremer 2015). Participation in additional after school programs increased future employment rates and reduced welfare receipts in adults who participated. It was found to have effects on long run fertility and labor outcomes as evidenced when schooling was subsidized through conditional cash transfers (Barham, Macours, and Maluccio 2024). These results imply that underinvestment in children's education is costly not only for the child's future self but also for aggregate productivity, employment and societal well-being.

The goal of this thesis is to understand how parental aspirations influence child education investment. Parental aspirations are viewed as future reference targets that households aspire to beat in the future. Beating these targets gives satisfaction while falling short gives frustration. Families are left with a trade-off between consuming now and investing toward the future (investing in education) that will yield greater future payoffs. Preferences accurately capture feelings of satisfaction and frustration through a reference dependent utility function referenced around the aspiration target (Kahneman and Tversky 1979). Having set an aspiration target they wish to reach, households are left to decide how much to invest. For low aspiration targets, beating the reference target to gain satisfaction is almost a given and hence investment is low, moderate aspirations inspire investment as investment yields payoffs that will aid in beating the aspiration target but high aspirations discourage invest-

ment as they are seen as unachievable even with investment (Ray and Genicot 2017). Thus, investment in child education yields an inverted U shape in aspirations. In future sections, this hypothesis is rigorously outlined and explained.

Empirically, the thesis utilizes rich household survey data from the Indonesia Family Life Survey wave 5 (IFLS-5). The IFLS is a nationally representative longitudinal survey that collected detailed information on socioeconomic status, expenditures, and individual attitudes across Indonesian households. Crucially, IFLS-5 (conducted in 2014–2015) included a self-anchoring ladder question asking respondents to rate their current economic standing on a 1–6 scale, as well as where they will be in the future. Using these responses, the aspirations ratio (measure of aspirations) is constructed. The household's per-capita education expenditure is constructed by aggregating all educational expenditures in the last month (such as school fees, uniforms, books, tutoring, etc.). This serves as the measure of household investment in child education. The relationship between the aspirations ratio and education investment is tested using regression analysis. The core specification is a quadratic regression: regressing per-capita education expenditure on the aspirations ratio and its square, including necessary controls (household assets, parental education, number of children etc.). The inclusion of the square term allows us to detect nonlinearity. Finding the coefficient on the linear term positive and on the quadratic term negative, would confirm an inverted U shape. Among other tests and specifications, a sub-sample analysis focusing on households with moderate aspirations is also carried out. Households whose current ladder position is in the middle range (steps 3–5 out of 6) are chosen as they are likely to have aspirations in a range that is not too high. For this subsample, we estimate a simpler linear regression of education spending on the aspirations ratio (plus controls). This provides a check on whether moderate aspiration do seek to aspire.

The analysis yields clear evidence that aspirations influence household educational investments in Indonesia, in a manner consistent with the proposed inverted U relationship. In the full-sample quadratic model, the coefficient on the aspirations ratio (linear term) is positive and significant, while the coefficient on the squared aspirations ratio is negative and significant. Specifically, the estimated model suggests that education spending increases with the aspirations ratio up to a point and then reduces. The turning point is around a

ratio of 4. Beyond that level, further increases in the aspirations ratio are associated with lower education spending. The magnitude of the estimated effects is substantial. A one-unit increase in the aspirations ratio initially corresponds to about a 26 percent monthly education expenditure per capita increase but this effect diminishes and eventually reverses as the ratio grows, reaching a turning point around 4. In the restricted sub-sample of moderate households (those on ladder steps 3–5), we find a simpler positive linear relationship between aspirations and investment. The data show that higher aspirations go hand-in-hand with greater education spending. The estimated linear effect is on the order of a 9 percent increase in per capita education expenditure if the aspirations target happens to increase by rung for individuals identifying on step 3. This aligns with the idea that within a reasonable range of aspirations, more aspiration drives more investment.

This thesis develops an original measure of capturing aspirations and provides crucial evidence to existing frameworks on aspirations (Ray and Genicot 2017; Dalton, Ghosal, and Mani 2016). It also contributes to the growing body of evidence on the impact aspirations has on key variables (Ross 2019). The important role of parents in shaping child outcomes is also highlighted and the evidence in this thesis complements research that found that mother's aspirations are strongly attached to child grade achievement (Serneels and Dercon 2021). The rest of the thesis is designed as follows; section 2 gives a brief background of Indonesia, section 3 is a detailed literature review, section 4 outlines the theoretical model and the testable implication, section 5 outlines the dataset and data preparation done, section 6 shows the main estimation strategy and results, section 7 shows additional robustness checks and section 8 concludes this thesis.

2 Background

Indonesia, the largest state in Southeast Asia, is an extremely diverse island country, reflected by its national motto "Unity in Diversity". In 2010, it became the world's fourth most populous country after China, India and the United States, with 237.6 million people making up about 3.5% of the world's population (Ananta, Arifin, and Hasbullah 2015). Economic inequality remained relatively low during the high-growth New Order phase that lasted from

the 1970s to the 1990s. However, following the country's democratic transition in the late 1990s, inequality rose sharply during the first decade of the new millennium, reaching a record high in 2011 (Tadjoeddin 2013).

Reviewing the extensive literature on Indonesian education, it is clear that the country has made substantial progress according to many common indicators. However, the evidence also shows that the educational outcomes remain highly unequal. Significant disparities persist across socioeconomic groups in both school attendance and examination performance. These differences connected to increasingly unequal labor market outcomes are further reinforced by the globalization of labor markets and increasingly restrictive domestic regulations, which together contribute to large segments of the Indonesian workforce becoming trapped in low-skill, low-wage occupations (Hill 2021).

Indonesia's education system is among the largest in the world. As of 2014, it had more than 50 million students, 3 million teachers, and 300,000 schools. The public school system follows the widely used 6–3-3 structure: six years of primary school, three years of lower secondary school, and three years of upper secondary studies. Although 12 years of schooling are formally compulsory, actual completion rates fall short of this. In 2011, the net enrollment rate dropped from 77.71 percent in lower secondary education to 57.74 percent in upper secondary education, indicating a substantial drop in attendance after the age of 15 (ASEAN Secretariat 2014). A variety of factors contribute to this pattern. While public primary and lower secondary education is free, registration fees and other additional costs remain. These were significantly reduced after 1998 but not completely eliminated, which may still pose financial barriers for lower-income households (Rosser and Joshi 2013). Concerns about educational quality further increase these issues: an estimated 55 percent of Indonesians who complete formal schooling are considered functionally illiterate (Barro and Lee 2013). Taken together, these features make Indonesia a compelling setting for studying differences in educational investments and outcomes. Its socioeconomic diversity, persistent inequality, and distinct institutional context generate substantial variation in access, attainment, and returns to education.

3 Literature Review

Aspiring to understand why households invest differently in their children's education remains the goal. Recent work on human capital investment shows that parents' decisions depend on how they perceive the economic returns to education relative to its opportunity costs (Jensen 2010). Evidence shows that updating beliefs about the returns to schooling can lead to substantial increases in enrollment. Complementary research highlights the role of school inputs and teacher quality in shaping learning outcomes and children's willingness to attend school (Chetty, Friedman, and Rockoff 2014). Together, this frames education investment and attainment as a rational response to information and constraints. Another important line of research focuses on how household constraints impact education investment. Parental beliefs about their children's academic ability, for example, have been shown to strongly affect allocation of resources within the household. Parents adjust their educational spending accordingly when provided with more accurate information about their child's performance (Dizon-Ross 2019).

While these frameworks highlight the structural and informational frictions that shape schooling decisions, they do not fully account for why households facing similar constraints and opportunities often invest in their children's education at dramatically different levels. This limitation has motivated a growing literature in behavioral and psychosocial explanations of educational investment, particularly within behavioral economics. Research shows that cognitive burdens can impair households' abilities to process information and plan ahead. This is because poverty and lack of time causes individuals to make poorer decisions as these conditions impose a cognitive load that reduces attention and effort (Mani et al. 2013). Such cognitive constraints could spill over into educational decision-making, leading disadvantaged households to make suboptimal or short-sighted investments in their children's schooling.

Recently, explanations of some of these differences in investment choices through the lens of aspirations have been developed. The earliest work on this highlights "the capacity to aspire" and comes from Sociology. This capacity is hypothesized to be a cultural and social capability that influences how individuals map out their futures and view it (Appadurai 2004). This is also supported by empirical evidence. For instance, it was found that ben-

eficiary parents of a Mexican antipoverty program had higher educational aspirations for their children, seen in a 15 percent increase in the proportion of parents who aspire for their children to finish college (Chiapa, Garrido, and Prina 2012). Building on this, theoretical models define aspirations as reference points that influence behavior. Founded in Prospect Theory (Kahneman and Tversky 1979), aspirations are seen as reference targets that individuals aspire to reach in the future. If reached, they gain satisfaction, but when falling short, they are left with frustration. The reference targets and the resulting emotions further shape their preferences and decision making. This subsequently impacts outcomes such as child education investment (Ray 2006; Ray and Genicot 2017; Genicot and Ray 2020). They hypothesize that aspirations do not linearly impact investment. They rather have an inverted U relationship, with excessive aspirations discouraging individuals from investing due to a fear of "aspirations failure", where the reference target is seen as unreachable.

In empirical research, numerous works on the key impact of aspirations have been conducted. For instance, experimental evidence shows that exposure to successful female leaders changes girls' aspirations and subsequently increases their educational attainment (Beaman et al. 2012). Further studies on rural Ethiopia reveal heterogeneity in parental aspirations by gender. Exposure to documentaries on female and male role models increased parental aspirations as well as enrollment rates, demonstrating that aspiration shifts can meaningfully impact educational outcomes (Bernard et al. 2019). In a study on financial investments of people in Nepal, a correlation between aspirations and future-oriented behavior was found, but only up to a certain point. Further, although aspirations might be a social phenomenon, they are found to either stimulate development or reinforce poverty (Janzen, Magnan, and Sun 2017). In a direct empirical test, it is observed that higher income aspirations reduce people's utility, ceteris paribus. Using individual level data on reported life satisfaction as a proxy measure for utility and income evaluation measures as proxies for people's aspiration levels, it was found that aspirations increase with people's income as well as with the average income in the community they live in (Stutzer 2003).

Extensive research was made on aspirations gaps and ratios, the focus of this work. For example, empirical evidence of a nonlinear relationship between the size of an adolescent's aspirations gap and human capital as a young adult was found. The adolescent's aspirations

gap in that study is quantified as the difference between the wage associated with the occupation they aspire and the wage associated with their parents' occupation. Higher aspirations at age 12 are correlated with higher human capital levels at age 19, but only up to a limit. Above this point, the relationship plateaus and human capital levels no longer increase. The children with the largest aspirations gaps have lower human capital levels than those with more moderate gaps. This resembles the inverted U relationship, which will be explored in this work, although in a different context (Ross 2019). This is also supported by evidence from India that reveals that that chronically poor households exhibit lower educational investments partly because of their aspirations. It is also found that a mother's aspirations are strongly attached to the child's grade achieved at 18. Furthermore, this relationship is nonlinear, suggesting the existence of a threshold and further reinforcing the impact of "aspirations failure", mentioned above (Serneels and Dercon 2021). These empirical insights confirm that aspirations are not merely reflective of constraints but can themselves act as independent determinants of investment behavior.

There have been few other attempts to build frameworks with aspirations as a key factor. In one such economic model, hope is understood as a function of aspirations, agency and pathways. It illustrates the vital role hope can play in the realization of causal effects from development interventions and how these emerge from the impact and interaction of three constituent elements of hope (Lybbert and Wydick 2018). Works on "aspirations failure" emphasize that households with limited role models or restricted opportunity sets may form low or stagnant aspirations that perpetuate underinvestment. It is stressed that "aspirations failure" is not a cause for poverty but rather a consequence of it and there exists a way by which just altering aspirations can offer a way out of a poverty trap (Dalton, Ghosal, and Mani 2016). Together, these studies suggest that a full understanding of household investment in children's education requires integrating structural factors with behavioral determinants.

4 Theoretical Model

4.1 The Budget Constraint

Following the work of Ray and Genicot (Ray and Genicot 2017; Genicot and Ray 2020), from a behavioral perspective, the goal is to study the implications that aspirations have on decision making, particularly investment decisions regarding child education. In the model of this work, aspirations are assumed to be scalars, each household has a current income $y > 0$ and in a single period, the household chooses to allocate this income between consumption c and investment k. This will then yield $z = f(k)$ in the next period. The budget constraint is given by

$$y = c + k \quad \text{with} \quad k \geq 0,\ f'(k) > 0 \text{ and } f''(k) \leq 0. \tag{1}$$

4.2 Aspirations and Preferences

Let a be the target that the household aspires to reach in the next period. A reference dependent utility function around a is defined. The utility of the household depends on three factors:

- current consumption c,
- the outcome of the next period $f(k)$,
- and a gain-loss utility term referenced around a which is the aspirational component that rewards exceeding a and penalizes falling short of a:

$$U(c,\, k,\, a) = u(c) + \delta \left[w_0 \left(f(k) \right) + \pi_G W_G \left(\left(f(k) - a \right)_+ \right) - \pi_L W_L \left(\left(a - f(k) \right)_+ \right) \right] . \tag{2}$$

Here, u, w_0, W_G, W_L are all increasing concave functions, $\delta > 0$ is the weight how much we value future outcomes today and π_G, $\pi_L \geq 0$ gives us the strength of gains and the strength of losses, respectively, where $(x)_+ = \max\{x,\, 0\}$. The intuition behind such a utility function follows from the fact that households set a target for themselves based on their view of society, and then evaluate their future outcomes based on this target. They gain a sense

of satisfaction by exceeding this target and conversely a sense of disappointment by falling behind. Frustration usually weighs stronger on the individual (Tversky and Kahneman 1991).

4.3 Solving

Based on (Ray and Genicot 2017), the "common elasticity growth model" is adapted to simplify the model. To start, an "AK" setting is imposed making the production function linear as it is argued that this formulation also allows for exogenous labor income in every period. This formulation implies $f(k) = \rho k = z$ with $\rho > 1$. This means, investing k yields ρk in the next period. Further, it is assumed that utilities are the same constant elasticity:

$$u(c) = c^{1-\sigma}, \quad w_0(z) = \delta z^{1-\sigma}, \quad W_G(e) = \delta \pi_G e^{1-\sigma}, \quad W_L(e) = \delta \pi_L e^{1-\sigma} \quad \text{with} \quad \sigma \in (0,\, 1). \tag{3}$$

Here, σ is the common curvature, with a lower value implying more elastic responses. $\delta > 0$ is the discount weight on next period's payoffs as viewed today. $\pi_G > 0$ is the weight on the additional utility boost received from achieving the aspiration target and $\pi_L > 0$ is the negative utility penalty received for falling short of it. Typically, $\pi_L > \pi_G$ due to loss aversion. $e \geq 0$ denotes the gap with $e = (z - a)_+$ and $e = (a - z)_+$ for the gain and loss, respectively, and $(x)_+ = \max\{x,\, 0\}$. Finally, the growth factor is given by $g = z/y$, the aspirations ratio is $r = a/y$, and the investment share is $i = k/y = g/\rho$. After imposing all these conditions on the utility function for the household in eq. (2), the gain-loss utility term is reduced to

$$\tilde{U}(g,\, r) = \left(1 - \tfrac{g}{\rho}\right)^{1-\sigma} + \delta \left[g^{1-\sigma} + \pi_G \, (g - r)_+^{1-\sigma} - \pi_L \, (r - g)_+^{1-\sigma}\right] \quad \text{with} \quad g \in [0,\, \rho]. \tag{4}$$

Once the household chooses an aspiration target a, their aspirations ratio r is fixed and they choose their level of investment k which yields z to maximize their utility function through their growth rate g. Since the utility function is piecewise, it is partitioned it into 2 zones, the satisfied and frustrated zones based on z. In the satisfied zone, the chosen k results in $g \geq r$, while in the frustrated zone, it is $g < r$. This gives rise to two First Order Conditions

(FOCs):

$$\textbf{Satisfied } (g \geq r): \quad \left(1 - \tfrac{g}{\rho}\right)^{-\sigma} = \delta\rho \left[g^{-\sigma} + \pi_G (g - r)^{-\sigma}\right], \tag{S_g}$$

$$\textbf{Frustrated } (g < r): \quad \left(1 - \tfrac{g}{\rho}\right)^{-\sigma} = \delta\rho \left[g^{-\sigma} + \pi_L (r - g)^{-\sigma}\right]. \tag{F_g}$$

4.4 Testable Implications

There exists an $r^* > 1$, such that $g^*(r)$ solves the satisfied FOC (S_g) for $r < r^*$, and the frustrated FOC (F_g) for $r > r^*$. A unique $r^* > 1$ exists by design of our utility function eq. (2). Arguing for the existence intuitively, first imagine r is low. That would mean the gain region of the reference dependent part of the utility function is active. This implies slight increases in the aspiration target still encourage investment. Investment (and hence $g^*(r)$) is increasing in aspirations in this region. Conversely, imagine r is low. That would mean the loss region of the reference dependent part of the utility function is active. If there is a slight increase in r, investing more yields negative utility in this loss region and households would choose to increase consumption instead. Thus investment (and $g^*(r)$) in aspirations is decreasing in this region. Since the optimal investment response moves up with r when aspirations are low, and moves down with r when aspirations are high, and because the optimal choice changes smoothly with r , there must be some intermediate aspiration level r^* at which the response switches from "increasing" to "decreasing". The strong assumptions made regarding their continuity and concavity ensure this. At point r^*, the household is exactly on the boundary between being satisfied and being frustrated, and this is the r^* that generates the inverted U in the aspirations ratio and is the point where investment is maximum. Thus, there are 3 conditions that are met. First, there exists a unique $r^* > 1$ at which the household is exactly on the boundary between the satisfied and frustrated regions, so that the optimal choice $g^*(r^*)$ just meets the target. Second, for aspirations below this threshold ($r < r^*$), the gain region is active and the optimal growth choice $g^*(r)$ (and hence investment) is increasing in r. Third, for aspirations above this threshold ($r > r^*$), the loss region is active and the optimal growth choice $g^*(r)$ (and investment) is decreasing in r. This yields our testable implication as these 3 conditions mean the following: Investment (keeping

income fixed) yields an inverted U in aspirations ratio. The graphical representation of this implication can be found in fig. 1.

5 Data Preparation

5.1 Demographics

The “Indonesia Family Life Survey" (IFLS-5) is the fifth wave of one of the longest-running and most comprehensive household panel datasets in the developing world. Conducted in 2014–2015, it followed the original 1993 baseline sample. The survey follows individuals, households, and communities over time, capturing information on demographics, education, health, labor, consumption, assets, aspirations, social capital, and local public services. Historically, the IFLS was created to provide a representative panel for the 83% of Indonesia's population living in the major island groups: Java, Bali, Sumatra, Kalimantan, Sulawesi, and West Nusa Tenggara. The sampling frame includes 13 provinces, chosen for their population size and diversity, and employs a stratified, multistage cluster design. By IFLS-5, the panel successfully connected with a high share of the original sample: recontact rates remained above 90% for surviving households, making IFLS one of the most stable longitudinal datasets globally.

At the community level, IFLS-5 surveys Enumeration Areas (EAs) as the smallest geographical unit. It spans 321 EAs, which are sample clusters. In the analyses of this work, the Standard Errors are clustered at the EA level. One issue of this definition is that these EAs were designed as sample clusters for round one of the IFLS survey and remained unchanged until the 5th round. Hence, in all of the following analyses, along with clustering Standard Errors at the EA level, fixed effects at Kecamatan are taken. Kecamatan is the third-level administrative sub-divison in Indonesia just below the regency or city level. In table 1, the distribution of households across these EAs and Kecamatans are shown.

With over 240 million Muslims, Indonesia is the largest Muslim nation in the world. Hence, it is natural that the sample mostly consists of households that are Muslim (over 90%). The religious split in the total sample can be seen in table 2. Nearly 50% of the households are Javanese, followed by Sundanese at around 10% and numerous other ethnicities contributing

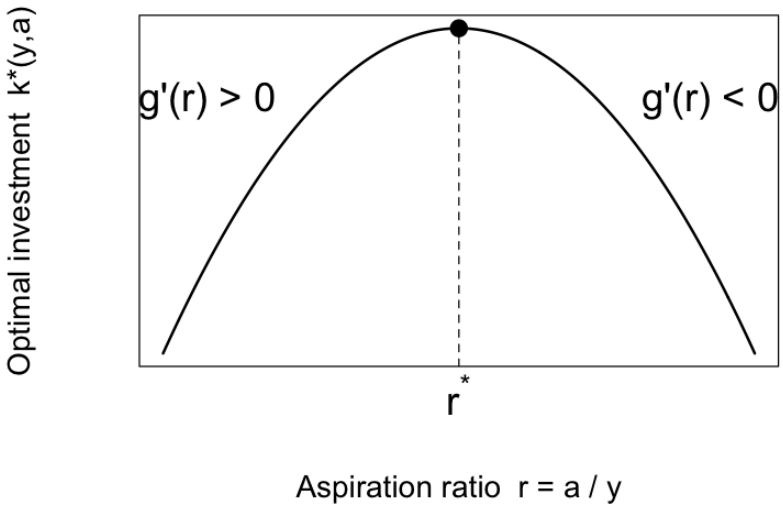


Figure 1: Inverted-U investment response to aspirations ratio.

Table 1: District and Enumeration Area (EA) characteristics.

	Number	Median households	Mean households	SD households
Kecamatan	87	16.00	53.67	94.83
EA	317	14.00	14.73	8.06

Notes: Households are counted using the main regression sample corresponding to table 6, column (1). For later analysis, clustering is done at the EA level and fixed effects at the Kecamatan level are taken.

small percentages each. This is in part due to the extremely large number of islands that Indonesia is composed of. In table A1, the ethnic distribution in the sample can be seen. This type of diversity is beneficial for the analysis as with more diversity, more certainty is gained that results are not confounded by other sociocultural practices.

5.2 The Aspirations Ratio

The main reason for choosing the IFLS 5 dataset is the presence of the Subjective Wellbeing module in the questionnaire. This module collects data regarding various behavioral topics namely mental health, risk aversion, present bias, life satisfaction and ambitions. Of these

Table 2: Distribution of household head religion.

Religion of head	Count	Share (%)
Islam	4,252	90.10
Hindu	208	4.45
Protestant	152	3.26
Catholic	54	1.16
Buddhist	2	0.04
Other	1	0.02
Total	4,669	100.00

Notes: Shares are expressed as percentages. Sample corresponds to the main regression sample in table 6, column (1).

various topics, the questions that are most of interest to this work are the ones where the individuals are asked to place themselves on an economic ladder. Specifically, participants needed to answer the question on which economic step they consider themselves that day based on a 1-6 ladder scale with 1 being the lowest step. Subsequently, they were asked on which economic step they will be five years from then with the same scale. These questions form the basis of the measure for aspirations in this work. Recalling the work of Ray and Genicot 2017, aspirations are defined as reference targets that individuals aspire to reach. These are heavily impacted by where they view themselves in the their society and compared to people around them. This is why, it is very difficult to extract these aspirational measures from existing datasets that do not explicitly aim at this. In this regard, these two questions prove extremely useful in measuring the “aspirations ratio" variable to verify the model in this work. Formally, the aspirations ratio is defined as

$$\text{aspirations ratio} = \frac{\text{economic step in the future}}{\text{economic step today}},$$

where the economic step in the future is the response the individual gave to the question “On which economic step will you be five years from now?" and the economic step today is the response the individual gave to the question “On which economic step do you consider yourself today?". This measure purely relies on the perception of the individual as to where

Table 3: Current economic step and aspirations ratio.

Current economic step	Median aspirations ratio	Mean aspirations ratio
1	2.000	2.581
2	1.500	1.551
3	1.333	1.359
4	1.250	1.221
5	1.200	1.106
6	1.000	1.000

Notes: The aspirations ratio is defined as the ratio of the household's economic step in the future to its current economic step. Sample corresponds to the main regression sample in table 6, column (1).

they view themselves in the society and where they would like to be as opposed to many other studies, which elicited aspirations through responses to preferred occupation choices, see for example Ross 2019. In table 3, the distribution of the aspirations ratio in terms of the economic step the households view themselves on today is shown. In this analysis, the aspirations ratio of the household head is used. It is also important to note that the aspirations ratio measure is very much impacted by the economic step individuals currently view themselves on by design. Only individuals on the lower end will have the capacity to have high aspirations ratios. Although this also translates to reality, where only poorer households have the ability to aspire to make large moves up the wealth ladders, this might later prove to be a limitation.

There is an immense rural vs. urban population split in Indonesia. Due to its large number of islands, it also consists of a huge number of rural households with different spending patterns. Of the 4669 households in the analysis sample, 1982 households are classified as rural and 2687 as urban. Table 4 shows the split in their current economic step between households in rural and in urban areas. They have very similar distributions and individuals largely sort themselves into similar economic distributions. This is rather interesting as one would expect urban households to rank themselves higher in general as they usually make have higher incomes reflected by the fact that in developing countries rural to urban migration is an extremely common phenomenon.

Table 4: Economic step today: rural vs. urban.

Economic step today (rural)	Frequency	Economic step today (urban)	Frequency
1	189	1	138
2	486	2	527
3	900	3	1,331
4	356	4	614
5	37	5	50
6	14	6	27
Total	1,982	Total	2,687

Notes: Sample corresponds to the main regression sample in table 6, column (1).

The distribution across economics steps today with regards to the aspirations ratio is also very similar in both rural and urban settings with urban households having marginally higher aspirations in the lower ladder bins. This can be seen in table 5. This means that both rural and urban populations view the economic ladder questions only in a local context. This fits into the literature on aspirations as it is assumed that individuals form aspirations through the view they have of the income distribution, which is heavily influenced by the society around them and how they are placed within that (Ray and Genicot 2017).

Table 5: Mean aspirations ratio for rural vs. urban by economic step today.

Economic step (rural)	Mean ratio	SD	Economic step (urban)	Mean ratio	SD
1	2.460	1.382	1	2.746	1.661
2	1.519	0.442	2	1.582	0.490
3	1.342	0.261	3	1.370	0.274
4	1.224	0.165	4	1.219	0.159
5	1.103	0.101	5	1.108	0.101
6	1.000	0.000	6	1.000	0.000

Notes: Sample corresponds to the main regression sample in table 6, column (1).
SD: Standard deviation.

5.3 The Dependent Variable

From our Testable Implications section, investments take an inverted U shape with respect to the aspirations ratio. As a measure of investment, the investment in the education of the

child is used (Ray and Genicot 2017; Genicot and Ray 2020). First, only children up to the age 15 as in the IFLS 5 questionnaire are considered. A reason for selecting this specific age is that, as mentioned before, dropout rates beyond the age of 15 become significantly larger (ASEAN Secretariat 2014). Hence, to construct child education investment, all money spent per child on education under the age of 15 in the last month is divided by the same number of children added up to get the value of the per capita child education investment (used interchangeably with per child education investment) for each household. A child is included only if they attended school in the last year. This variable (later treated in log or normalized) will be the main dependent variable in this analysis. The benefits of taking logs in such an analysis is that it tames skewness and outliers and helps when we compare across various income groups.

The median per child education expenditure in rural and urban households is vastly different. Urban households spend 50% more on education per child than rural households. The median per child education expenditure distribution by economic step today in urban households is also more as expected than in rural households. There is a steady increase from economic steps 1 to 5 and then a small drop in urban households, whereas in rural households, there is a small drop at step 5 followed by a great drop at step 6. These findings are visible in fig. 2. A possible reason is the small sample size at economic step 6. As can be seen in table 4, less than 1% of rural and urban households place themselves on step 6.

5.4 Other Controls

The economic step today is always controlled for, so that all variations in education expenditure that arise purely due to where household heads place themselves on the economic ladder should be absorbed. The urban vs. rural split is also controlled for by using a dummy variable due to the vast differences in expenditure seen between rural and urban households in fig. 2. Further controls are split into two categories: household characteristic controls and household head controls. For household characteristic controls, assets per capita is included, which is the total monetary value of all assets possessed by the household divided by the number of individuals in the household. Also included are the number of children, the number of children aged 6-12, the number of children aged 12-18, a dummy variable for whether

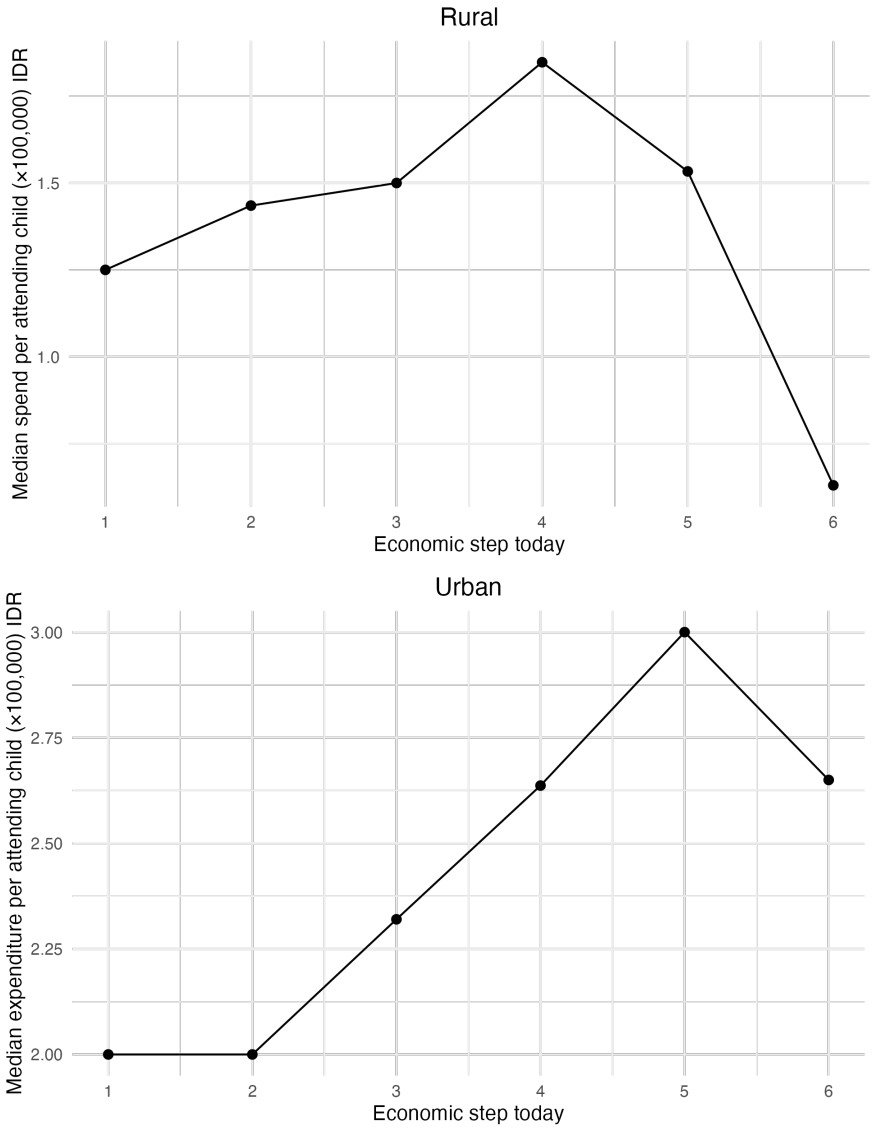


Figure 2: Median expenditure per attending child for rural vs. urban by economic step today.

it is a two-parent household, the monetary value of all the government assistance received towards schooling costs per school-going child as well as the dependency ratio for households, which is defined as follows:

$$\text{dependency ratio} = \frac{\text{sum of children and elderly}}{\text{number of working age adults}} .$$

Included in the household head controls are age, years of education of the household head and dummies for marriage status, religion and ethnicity of the household head. A detailed set of summary statistics for all these variables can be seen in table A3. To absorb geographical differences, standard errors are clustered at the Enumeration Areas (EAs) level and fixed effects are taken at the Kecamatan level. The EA, which is the smallest geographical unit, is the cluster that shares the same sample characteristics, but the EA units were frozen in round one of IFLS. Hence, households that moved away still share their original EA codes. To make the results more robust, fixed effects for Kecamatans, which are the next smallest geographical unit available in IFLS5, are taken. The distribution of households at the EA and Kecamatan level can be seen in table 1.

6 Estimation Strategy and Results

6.1 Main Specification

The goal is to test the implication that investment produces an inverted U shape with respect to the aspirations ratio with a straightforward quadratic regression specification. The main household specification is as follows:

$$y_h = \beta_0 + \beta_1 a_h + \beta_2 a_h^2 + \sum_{s=2}^{6} \beta_{3s}\{\text{step}_h = s\} + X_h'\theta + Z_h'\gamma + \alpha_{k(h)} + \varepsilon_h \,, \tag{5}$$

where

y_h is log of education expenditure per attending child for household h;

a_h is the aspirations ratio for household h;

a_h^2 is the squared aspirations ratio for household h;

$\text{step}_h = s$ is 1 if household h reports current economic step s on the ladder (step 1 omitted);

$X_h'\theta$ is a vector of household demographic controls (and their coefficients);

$Z_h'\gamma$ is a vector of household head controls (and their coefficients);

$\alpha_{k(h)}$ is a fixed effect for the Kecamatan k in which household h resides;

ε_h is the error term capturing unobserved determinants of y_h.

The vector of household demographic controls includes assets per capita, dependency ratio, number of children, number of children 6-12, number of children 12-18, amount of government assistance received per child towards education in logs and dummies for whether the household is an urban one and for whether it is a two-parent household. The vector of household head controls include the age of the household head, the years of education of the household head and dummies for the marriage status, religion and ethnicity of the household head. It is hard to argue that the results obtained from this specification are causal even if there are a rich set of controls, clustered standard errors and fixed effects. Reverse causality for example could be an issue as investment behaviour could very well impact aspirations. Although the inverted U is very unique and it is hard to imagine why our data would yield an inverted U if it were true that there was reverse causality, it is difficult to interpret these results as causal without employing a causal identification strategy. The lack of an explanation disproving causality does not imply causality.

6.2 Results and Interpretation of Main Specification

Table 6 presents the results of the main specification. The dependent variable y_h (education expenditure per attending child) is in log and scaled by 100. Column 1 of the table runs the specification with only the controls for the current economic step, the second column adds

the household demographic controls on top and column 3 is the full specification with the household head controls also included. The results are as expected for fitting an inverted U as the linear term has a positive coefficient and the square term has a negative coefficient implying that an increase in aspirations initially sees a raise in education spending, but beyond an aspirations ratio of roughly 4, further increases are associated with lower spending. Having the dependent variable y_h in $100\times \log(\cdot)$ is extremely useful as the coefficients on the right hand side can be directly interpreted as percentage increases to the dependent variable y_h. The main interest lies in column (3) as that is the full specification with all controls. The coefficients on the aspirations ratio and its square are $\beta_1 = 34.518$ and $\beta_2 = -4.334$. The marginal effect of the aspirations ratio a_h on education expenditure per child is

$$\frac{\partial y_h}{\partial a_h} = \beta_1 + 2\beta_2 a_h = 34.518 - 8.668\, a_h.$$

Thus, increasing the aspirations ratio from, for instance, 1 to 2 is associated with a raise in education spending of about 26% (evaluated at $a_h = 1$), from 2 to 3 with a rise of about 17%, and from 3 to 4 with a rise of about 9%, holding other variables fixed. The implied turning point of the quadratic is

$$a^* = -\frac{\beta_1}{2\beta_2} \approx \frac{34.518}{2 \times 4.334} \approx 4\,.$$

So, education expenditure per child increases with the aspirations ratio for $a_h < a^*$, but the marginal gains become smaller as a_h approaches a^*, and for $a_h > a^*$ further increases in aspirations are predicted to *reduce* education expenditure. Therefore, the positive linear term and negative squared term imply a statistically significant inverted U relationship between aspirations and education investment. The coefficients of the linear and squared term are interpreted as significant as they are jointly significant as evaluated using the Wald statistic in table A4. Of the control variables included, Assets per capita (millions) has a highly significant albeit small positive impact. Number of children aged 12-18, years of education of household head, the urban dummy and the marriage status of the head all have large, positive and significant impacts while the Two-parent household dummy has a large, significant and negative impact.

6.3 Normalised Education Expenditure

While the results from the main specification are promising, there is one persistent issue that needs solving. The distribution of the aspirations ratio is correlated with the ladder rank of where the individual identifies themselves on the economic ladder at present (table 3) and the per capita expenditure on education is also strongly correlated to this. Thus, it is beneficial to look at a normalized version of education expenditure. In the normalised specification, the dependent variable from eq. (5) is altered to a normalised version of it. The new dependent variable for household h is defined as

$$y_h = 100 \log \left(\frac{\text{education expenditure per capita}_h}{\text{economic step today}_h} \right), \tag{6}$$

where education expenditure per capita is household h's education expenditure per attending child in the last month and economic step today$_h$ is the household's initial step on the aspirations ladder. Dividing education expenditure per child by economic step today$_h$ scales investment relative to the household's perceived current economic step, so that y_h measures education investment per child per unit of initial status, in percentage terms. This normalisation makes the dependent variable more comparable across households at different initial steps: a given percentage change in y_h then reflects a change in education spending *relative to where the household starts on the ladder*, rather than in absolute rupiah terms alone. It also helps disentangle the role of the aspirations ratio from simple level differences in perceived status. All other specifications in this model remain the same as in the main specification (eq. (5)) including all controls, standard error clustering and fixed effects.

Table 6: Impact of aspirations ratio on education expenditure.

	(1)	(2)	(3)
Aspirations ratio	35.390***	33.802***	34.518***
	(12.105)	(11.818)	(12.071)
(Aspirations ratio)^2	-3.948**	-4.141**	-4.334**
	(1.903)	(1.896)	(1.899)
Step today = 2	16.363	5.414	1.255
	(13.643)	(12.491)	(13.016)
Step today = 3	47.980***	28.134**	21.502
	(13.630)	(12.260)	(13.071)
Step today = 4	61.910***	31.757**	22.868*
	(14.433)	(13.067)	(13.758)
Step today = 5	56.339**	27.804	21.154
	(22.057)	(20.634)	(20.953)
Step today = 6	14.633	-20.965	-27.949
	(49.825)	(50.396)	(49.828)
Assets per capita (millions)		0.278***	0.260***
		(0.034)	(0.032)
Dependency ratio		3.140	0.585
		(7.056)	(7.114)
Number of children		-0.999	-0.366
		(4.751)	(4.662)
Number of children aged 6–12		1.717	1.893
		(5.245)	(5.266)
Number of children aged 12–18		10.716***	11.073***
		(3.601)	(3.541)
Two-parent household dummy		-2.491	-20.098***
		(7.721)	(6.826)
log(Govt. assistance per child)		-0.627	-0.535
		(0.389)	(0.382)
Urban dummy		42.380***	38.460***
		(5.348)	(5.288)
Age of household head			-0.083
			(0.278)
Years of education of household head			1.084***
			(0.373)
Head marriage status dummy			32.583***
			(11.758)
Religion of head dummies	No	No	Yes
Ethnicity of head dummies	No	No	Yes
Kecamatan FE	Yes	Yes	Yes
S.E.: Clustered	EA	EA	EA
R^2	0.04078	0.07421	0.08852
Observations	4,669	4,651	4,651

Notes: Dependent variable is $100 \times \log$(education expenditure per capita). Assets per capita measured in millions. Standard errors in parentheses. *, **, *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

Table 7: Impact of aspirations ratio on normalised education expenditure.

	(1)	(2)	(3)
Aspirations ratio	33.938***	32.406***	33.337***
	(11.694)	(11.382)	(11.575)
(Aspirations ratio)^2	-3.754**	-3.953**	-4.164**
	(1.852)	(1.842)	(1.838)
Step today = 2	-52.183***	-63.073***	-66.819***
	(13.552)	(12.389)	(12.891)
Step today = 3	-61.478***	-81.222***	-87.396***
	(13.573)	(12.186)	(12.946)
Step today = 4	-76.069***	-106.036***	-114.296***
	(14.294)	(12.882)	(13.500)
Step today = 5	-103.311***	-131.671***	-137.624***
	(20.825)	(19.309)	(19.614)
Step today = 6	-156.521***	-191.218***	-197.734***
	(44.636)	(44.708)	(44.122)
Assets per capita (millions)		0.273***	0.255***
		(0.033)	(0.031)
Dependency ratio		3.538	1.252
		(6.452)	(6.485)
Number of children		-0.960	-0.364
		(4.448)	(4.358)
Number of children aged 6–12		0.547	0.668
		(4.956)	(4.971)
Number of children aged 12–18		10.795***	11.027***
		(3.410)	(3.351)
Two-parent household dummy		-2.610	-18.726***
		(7.245)	(6.550)
log(Govt. assistance per child)		-0.658*	-0.569
		(0.374)	(0.367)
Urban dummy		42.515***	38.466***
		(5.189)	(5.115)
Age of household head			-0.046
			(0.259)
Years of education of household head			1.078***
			(0.352)
Head marriage status dummy			29.991***
			(11.095)
Religion of head dummies	No	No	Yes
Ethnicity of head dummies	No	No	Yes
Kecamatan FE	Yes	Yes	Yes
S.E.: Clustered	EA	EA	EA
R^2	0.05635	0.09475	0.10933
Observations	4,669	4,651	4,651

Notes: Dependent variable is 100×log(normalised education expenditure per capita). Assets per capita measured in millions. Standard errors in parentheses. *, **, *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

6.4 Results and Interpretaion of Normalised Specification

Table 7 presents the results using our new Dependent Variable. We focus on the results in the full specification in column (3). The coefficients on the aspirations ratio and its square are $\beta_1 = 33.337$ and $\beta_2 = -4.164$, with dependent variable being the one we just defined (Equation (6)). When using *normlaised* per capita expenditure is used, caution should be exercised when interpreting coefficients. Since y_h is in $100 \times \log(\cdot)$ units, these coefficients can be read as approximate percentage effects on *normalised* education expenditure per child. The marginal effect of the aspirations ratio a_h on y_h is

$$\frac{\partial y_h}{\partial a_h} = \beta_1 + 2\beta_2 a_h = 33.337 - 8.328\, a_h\,.$$

Evaluated at different aspirations ratios, this implies that increasing a_h from 1 to 2 is associated with roughly a 25% rise in normalised education spending (marginal effect at Economic step today=1), from 2 to 3 with about a 16.7% rise, and from 3 to 4 with about an 8.4% rise, holding other covariates fixed. These marginal effects will clearly vary by Economic step today with the marginal effects being inversely proportional to the Economic step today. That is, increasing a_h from 1 to 2 for example sees a higher rise in education spending for individuals in the lower Economic step today, than for the ones with higher Economic step today. The implied turning point of the quadratic is similar to the main specification at

$$a^* = -\frac{\beta_1}{2\beta_2} \approx \frac{33.337}{2 \times 4.164} \approx 4\,,$$

so for $a_h < a^*$ higher aspirations raise education expenditure per child relative to initial step, but the marginal gains shrink as a_h approaches a^*, and for $a_h > a^*$ further increases in aspirations are predicted to reduce normalised education spending. Thus, even for the normalised dependent variable, the positive linear term and negative squared term imply a clear inverted U relationship between the aspirations ratio and education investment relative to initial status. The coefficients on the linear and square term are statistically significant individually but are also jointly significant as can be seen in Table A4. Among controls, richer households (higher assets per capita)have significantly higher normalised education expen-

diture per capita, and households with more children aged 12–18 also spend more. Urban households spend substantially more than rural ones, indicating large urban vs rural gaps in education spending. Finally, households whose head has more years of schooling, and those with a married head, have significantly higher normalised education expenditure per capita.

6.5 Revisiting the Urban vs. Rural Divide

One might be reminded of the vast differences between households identified in Rural vs Urban households in Indonesia. Although they sort themselves into similar Economic ladder placements (Table 4) and also have a very similar distribution of Aspirations Ratios across Economic ladder placements (Table 5), they have vastly different education expenditure patterns (**??**). The expenditure patterns of Rural households does not see as steady an increase across current Economic ladder placements as is visible from Urban households and in fact even sees a decrease at the higher ladder placements. This calls into question as to how well the hypothesis might apply to Rural households. Firstly, education attainment levels and importance in Urban areas are far higher and individuals stay much longer in school in Urban areas. Individuals who migrated to the city as children attained three more years of education, compared to observably similar individuals who remained in rural areas (Resosudarmo and Suryadarma 2014). For these reasons, I split my sample into a Rural only and Urban only sample. The results are as expected. Evidence from Urban only sample (Table 8) strongly supports our model whereas the evidence from the Rural only sample does not (Table A2).

Table 8 restricts the sample to urban households and shows that the relationship between the Aspirations Ratio and Education expenditure per capita is nonlinear. The 1st column refers to the dependent variable from our main specification (Equation (5)), the 2nd refers to the dependent variable from our Normalised specification (Equation (6)) and the 3rd column is the same as the main specification (Equation (5)) but using a restricted subsample which only includes households who answered the ladder question "On which Economic step do you consider yourself today?" with $\{1, 2, 3\}$. Across all three columns, the coefficient on the aspirations ratio is positive and significant (e.g. 34.466 in col. (1), 36.468 in col. (3)), while

the squared term is negative and significant (e.g. -5.677 in col. (1), -5.640 in col. (3)), implying an inverted–U relationship. With the dependent variable defined as $100\log(1+\cdot)$, these coefficients can be interpreted as approximate percentage effects: in column (1), the marginal effect of the aspirations ratio a on Education spending per capita is

$$\frac{\partial y}{\partial a} = 34.466 - 11.354\, a\,,$$

so increasing a from 1 to 2 is associated with a rise in education spending on the order of 20–25%, but the marginal gains shrink as a rises and become close to zero around $a^* \approx 3$, beyond which further increases in aspirations are predicted to *reduce* education expenditure per child. A very similar pattern holds when the dependent variable is normalised by the initial step (col. (2)) and in the subsample with initial steps $\{1, 2, 3\}$ (col. (3)), indicating that the inverted U is robust to both normalisation and sample restriction for Urban households. The Wald test for the joint significance of the linear and squared term also yield significant results (Table A4). All regressions include Kecamatan fixed effects and cluster Standard Errors at the EA level.

The Rural households only subsample on the other hand does not yield very significant results. There is some weak significance for the existence of a linear relationship between the Aspirations Ratio across using the main specification (Equation (5)) and the normalised specification (Equation (6)). These results are viewed keeping in mind that Rural areas in Indonesia lag behind in school and teacher infrastructure among numerous other issues. A survey was conducted in 270 rural and remote primary schools between 2016-2017 and survey respondents included principals, teachers, students, school committees, parents, and village heads. Most students tested were performing two grade levels below their current grades. In terms of teacher infrastructure in Rural areas, surveyed schools and villages faced connectivity challenges that may discourage the best teachers from working in these areas. They are on average 149 km (92.5 miles) or five hours away from district capitals. Only around 29 percent on average are connected to the power grid and only 17 percent have internet access. The second issue is teacher qualifications, teacher composition, and the necessity for multi-grade teaching. Thirty-four percent of teachers and 18 percent of principals only have

high school degrees. Permanent civil-servant teachers only constitute 40 percent of the teaching force, with the shortfall being filled by short-term, contracted teachers (World Bank 2019).

7 Robustness Checks

7.1 Subsampling

In this section, the impact on the results is seen by considering various sub-samples. As discussed earlier and by looking at the distribution of the Aspirations ratio, it is seen that the Aspirations ratio is distributed very differently across each step in the Economic step ladder individuals place themselves table 5. Although these differential effects are controlled for by adding dummies for each Economic step, there are numerous benefits to viewing various sub-samples of our dataset. For one, the number of households identifying themselves in the higher steps of the Economic ladder are far fewer than than ones identifying themselves around the middle or lower steps. By design, these higher households can only have moderate aspirations as they are already on a high step of the Economic ladder and since they are on a high step, it is natural that these households have high Education expenditure. Thus for more comparable results, considering a subsample of households in the lower rungs of the Economic ladder proves useful.

The first 2 sub-samples looked at are by Economic step today. The summary statistics for variables in these sub-samples are found in tables A5 and A6 The first sub-sample only includes households that answered $1, 2$ or 3 to the question asking "On which Economic step do you consider yourself today?" and the second subsample only includes those that answered 1 or 2. The main specification is used for this subsample (eq. (5)). To recap, the main household specification for the sub sample is $[n, m]$ by current Economic step is given by;

$$y_h = \beta_0 + \beta_1 a_h + \beta_2 a_h^2 + \sum_{s=n}^{m} \beta_{3s}\{\text{step}_h = s\} + X_h'\theta + Z_h'\gamma + \alpha_{k(h)} + \varepsilon_h \,, \tag{7}$$

where

y_h is log of education expenditure per attending child for household h;

a_h is the aspirations ratio for household h;

a_h^2 is the squared aspirations ratio for household h;

$\text{step}_h = s$ is 1 if household h reports current economic step s on the ladder (step 1 omitted);

$X_h'\theta$ is a vector of household demographic controls (and their coefficients);

$Z_h'\gamma$ is a vector of household head controls (and their coefficients);

$\alpha_{k(h)}$ is a fixed effect for the Kecamatan k in which household h resides;

ε_h is the error term capturing unobserved determinants of y_h.

The vector of household demographic controls includes Assets per capita, Dependency ratio, Number of children, Number of children 6-12, Number of children 12-18, amount of Govt assistance received per child toward education in logs and dummies for whether the household is an Urban one and for whether it is a 2 parent household. The vector of household head controls include the age of the household head, the years of education of the household head and a dummy for the marriage status.

The sub-sample results in table 9 show that the impact of the aspirations ratio on education expenditure per capita is an inverted U. In both columns, the aspirations ratio enters positively and significantly, while its square is negative and significant. The Wald test for the joint significance of the linear and quadratic term are also significant for both subsamples (table A4). The coefficients (col. (1): 33.091 and -4.013; col. (2): 34.547 and -4.118), imply that education spending rises with aspirations at low levels but eventually peaks and declines. With the dependent variable defined as $100\log(\cdot)$, these coefficients can be read as approximate percentage changes. For households starting on steps $\{1, 2, 3\}$, the marginal effect of the aspirations ratio a is $\partial y/\partial a \approx 33.091 - 8.026\,a$, so increases in a from 1 to 2 are associated with sizeable gains in education expenditure, but the marginal return shrinks and turns negative around $a \approx 4.1$, consistent with results from previous specifications. A similar turning point obtains in the $\{1, 2\}$ subsample as well further reinforcing results.

Among controls, assets per capita are positively associated with education expenditure (richer households spend more), higher numbers of older children (12–18) are linked to greater spending, and the urban dummy is large and highly significant in both columns, indicating substantially higher education expenditure per capita in urban areas even within these restricted initial step sub-samples. In contrast, the dependency ratio, total number of children, and government assistance per child are not significant once other covariates and Kecamatan fixed effects are included. Overall, these sub-sample results confirm that the inverted U relationship between aspirations and education expenditure is present even when restricting our sample to poorer households starting on the lowest rungs of the economic ladder and are not driven purely by outliers.

Looking back at all our models thus far, the kink or the turning point for our model is set at an Aspirations Ratio level of $\approx 3-4$. These are extremely high levels of Aspirations Ratio that can only be reached by individuals for whom Economic step today $= 1$. Individuals who select Economic step today at the higher steps on the ladder can only have moderate aspirations and never reach beyond the kink. Thus the analysis is replicated using the main specification eq. (7) by restricting the sample to households that identify on Economic step today $= 2$. This step is chosen in particular as firstly, around 20 percent of all households lie on this step which is much higher than Economic step today $= 1$ and a sufficient number of households is needed. Next, households on this step can still have a wider range of aspirations compared to steps above them as their aspirations ratio can go up to 3.

The specification in table 10 uses $100 \times \log(1 + \text{Education expenditure per capita})$ as the dependent variable and the coefficients can thus be interpreted directly as percentage changes in the dependent variable. The linear coefficient on the aspirations ratio is large and positive (90.77), while the squared term is negative (-16.67). Both estimated with relatively large standard errors so that neither is individually significant at conventional levels. However, a Wald test (found in table A4) of the joint null $H_0 : \beta_{\text{dm}} = \beta_{\text{dm}^2} = 0$ is rejected, indicating that the two coefficients are jointly significant and that the nonlinear (inverted U) relationship between aspirations and education expenditure remains relevant even within this narrow Economic step today $= 2$ subsample. The implied marginal effect of aspirations is $\partial y/\partial a \approx 90.77 - 33.34\, a$, so education spending is predicted to increase with the aspirations

ratio at low values of a but to peak and then decline around $a \approx 2.7$. Among the controls, assets per capita are positively and significantly associated with education expenditure (urban households with higher assets spend more per child), the urban dummy is large and highly significant, and having a married household head is positively associated with education spending at the 10 percent level. Other household composition and assistance variables are imprecisely estimated once Kecamatan fixed effects and the full set of controls are included.

7.2 The Moderate Aspirer

More than 50 percent of the households used in the main specification sample (table A3), are from the subset of ones that identify themselves on the Economic step today ladder question at levels $3, 4$ and 5 (summary statistics found in table A7). This sub-set of people are among the ones that spend the highest on education (per capita) (fig. 2) and by design these households cannot have extremely high aspirations as they are already on the higher end of the distribution. How does aspirations impact the investment choices of these households? Since these households are already on the higher end of the distribution, it is expected that these households do not experience "aspirations failure"; the scenario where the reference target they aspire to reach based on their view of the society around them is seen unrealistically and proves as a demotivating factor rather than a motivating one. Hence, based on eq. (2) it is anticipated that all of these households will lie in the "Satisfied zone" where $\partial g/\partial r$ is positive meaning investment is increasing aspirations ratio. To test this, a linear regression specification is implemented. The specification is as follows:

$$y_h = \beta_0 + \beta_1 a_h + \sum_{s=n}^{m} \beta_{3s}\{\text{step}_h = s\} + X_h'\theta + Z_h'\gamma + \alpha_{k(h)} + \varepsilon_h \,, \tag{8}$$

where

y_h is log of education expenditure per attending child for household h;

a_h is the aspirations ratio for household h;

$\text{step}_h = s$ is 1 if household h reports current economic step s on the ladder (step 1 omitted);

$X_h'\theta$ is a vector of household demographic controls (and their coefficients);

$Z_h'\gamma$ is a vector of household head controls (and their coefficients);

$\alpha_{k(h)}$ is a fixed effect for the Kecamatan k in which household h resides;

ε_h is the error term capturing unobserved determinants of y_h.

In table 11, the results of households that identify themselves on the Economic step today ladder question at levels $3, 4$ and 5 are found. The coefficient on the Aspirations ratio is positive and statistically significant. Since the dependent variable is $100 \times \log(\cdot)$ and the coefficients can directly be interpreted as percentage increases, a one-unit increase in the ratio is associated with roughly a 27–28 percentage increase in education expenditure per capita, holding all other co-variates constant. For example, if a household on Economic step 3 aspires to move to step 4 rather than staying on step 3, their education expenditure per capita would go up by around 9 percentage. The step dummies corresponding to the current Economic step of the household today are small and imprecisely estimated, indicating no clear level differences in spending across this range of initial steps once aspirations and controls are accounted for. Among controls, higher per-capita assets are strongly associated with greater education expenditure, the number of children aged 12–18 is positively related to spending, and urban households spend markedly more than rural households. Two-parent households spend somewhat lesser, and the years of education of the household head is positively associated with education expenditure, while most other controls are not precisely estimated in this subsample. The results and robustness check presented in thesis are in accordance with the literature. Using the original measure of aspirations, evidence for the existence of an inverted U relationship between investment and aspirations is found. Evidence for the existence of a satisfaction where increases in aspirations inspire increased investment is also found. These

findings are accordance with the literature on aspirations and lays the groundwork toward a future study on the causal impacts aspirations can have.

8 Conclusion

This thesis introduces an original way of quantifying aspirations and further verifying theoretical models on aspirations. The aspirations measure can used can be further bettered and improved especially using questionnaires that elicit responses that help in constructing an aspirations measure. This measure can also be constructed using questions regarding expectations or goals. For example, the India Human Development Survey includes questions on the education attainment a child wishes to achieve in the future (Desai, Vanneman, and National Council of Applied Economic Research, New Delhi 2019). Using this, a similar aspirations measure can be formulated to further study the impact it has. Understanding aspirations and how it impacts other outcomes could present a novel, cost-effective way of encouraging human capital investments. Aspirations failure leaves individuals stuck on a loop where it encourages them to invest lesser which will once again put them in their original position (Ray and Genicot 2017). For example, removing the phenomena of low-income children stuck attending school in low-income neighborhoods by shuffling the school system could drastically improve motivation levels. There are numerous other examples like this which could help shift an individual's reference target to induce motivation. With further research in this area, empirical studies and more innovation from the field of behavioral economics in helping understand decision making, inducing positive change by altering aspirations could prove to be the most cost effective way.

Table 8: Urban households subsample.

	Main	Normalised	Subsample
Aspirations ratio	34.466**	33.886**	36.468**
	(16.006)	(15.173)	(14.866)
(Aspirations ratio)^2	-5.677**	-5.560**	-5.640**
	(2.496)	(2.387)	(2.254)
Step today = 2	-26.541*	-94.425***	-16.200
	(15.393)	(15.108)	(14.514)
Step today = 3	-15.825	-123.730***	-2.454
	(14.426)	(14.134)	(13.771)
Step today = 4	-19.165	-155.134***	
	(16.805)	(16.190)	
Step today = 5	-2.824	-162.686***	
	(19.091)	(18.817)	
Step today = 6	-28.604	-199.986***	
	(55.767)	(49.862)	
Assets per capita (millions)	0.296***	0.289***	0.256***
	(0.037)	(0.036)	(0.051)
Dependency ratio	4.752	5.218	15.229
	(9.009)	(8.227)	(9.377)
Number of children	1.534	1.585	-1.785
	(5.899)	(5.482)	(7.099)
Number of children aged 6–12	0.999	-0.091	1.456
	(6.766)	(6.367)	(8.358)
Number of children aged 12–18	14.237***	13.951***	13.286**
	(4.557)	(4.267)	(5.537)
Two-parent household dummy	-14.815*	-13.676*	-13.494
	(8.305)	(7.986)	(9.847)
log(Govt. assistance received per child)	-0.954*	-0.978**	-0.918
	(0.496)	(0.470)	(0.582)
Age of household head	-0.342	-0.269	0.053
	(0.398)	(0.368)	(0.407)
Years of education of household head	0.755	0.781*	0.577
	(0.487)	(0.457)	(0.590)
Head marriage status dummy	33.009**	30.302**	38.172**
	(15.662)	(14.728)	(18.747)
Religion of head dummies	Yes	Yes	No
Ethnicity of head dummies	Yes	Yes	No
Kecamatan FE	Yes	Yes	Yes
S.E.: Clustered	EA	EA	EA
R^2	0.09849	0.12769	0.06544
Observations	2,682	2,682	1,992

Notes: Column 1 dependent variable follows from the main specification, column 2 from the normalised specification, and column 3 from the subsample specification. Dependent variable is in $100 \times \log(\cdot)$. Religion and ethnicity of head dummies are dropped in column (3) due to collinearity. Standard errors in parentheses. *, **, *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

Table 9: Modelling on subsamples based on initial economic step. Impact of aspirations ratio on education expenditure.

	(1) Initial step 1,2,3	(2) Initial step 1,2
Aspirations ratio	33.091***	34.547**
	(12.084)	(16.232)
(Aspirations ratio)^2	-4.013**	-4.118*
	(1.904)	(2.460)
Step today = 2	4.575	2.522
	(12.909)	(13.179)
Step today = 3	26.852**	
	(12.996)	
Assets per capita (millions)	0.212***	0.223*
	(0.044)	(0.126)
Dependency ratio	10.161	15.357
	(6.367)	(9.507)
Number of children	-4.468	-14.802*
	(5.220)	(8.653)
Number of Children aged 6–12	2.752	6.499
	(6.239)	(9.962)
Number of Children aged 12–18	10.280**	13.741*
	(4.087)	(7.682)
Two-parent household dummy	-15.383**	-23.681
	(7.315)	(14.883)
log(Govt. assistance received per child)	-0.602	-0.924
	(0.455)	(0.897)
Urban dummy	45.910***	60.063***
	(5.850)	(10.334)
Age of household head	0.214	0.568
	(0.292)	(0.475)
Years of education of household head	0.850*	1.163
	(0.441)	(0.938)
Head marriage status dummy	36.684***	55.049**
	(13.932)	(27.002)
Religion of head dummies	No	No
Ethnicity of head dummies	No	No
Fixed effects	Kecamatan	Kecamatan
S.E.: Clustered	EA	EA
R^2	0.06704	0.08721
Observations	3,556	1,329

Notes: Dependent variable is $100 \times \log$(Education expenditure per capita). Religion and Ethnicity dummies excluded from regressions (shown as 'No'). Assets per capita measured in millions. Standard errors in parentheses. *, **, *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

Table 10: Impact of aspirations ratio on education expenditure (economic step today = 2 only)

	(1)
Aspirations ratio	90.770
	(60.980)
(Aspirations ratio)^2	-16.670
	(15.730)
Assets per capita (millions)	0.3341**
	(0.1391)
Dependency ratio	17.080
	(13.540)
Number of children	-15.680
	(11.640)
Number of Children aged 6–12	8.304
	(12.650)
Number of Children aged 12–18	13.460
	(10.220)
Two-parent household dummy	-16.200
	(15.910)
log(Govt. assistance received per child)	-1.359
	(1.132)
Urban dummy	50.390***
	(12.080)
Age of household head	0.6807
	(0.6410)
Years of education of household head	1.001
	(1.058)
Head marriage status dummy	69.680*
	(39.030)
Kecamatan FE	Yes
S.E.: Clustered	EA
R^2	0.08102
Observations	1,005

Notes: Dependent variable is 100 × log(education expenditure per attending child). Sample restricted to households with Economic step today 2. *, **, *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively. A Wald test of the joint null that the coefficients on the aspirations ratio and its square are both zero is rejected at conventional significance levels (table A4).

Table 11: Linear impact of aspirations ratio on education expenditure on sub-sample (Economic step today $\in \{3, 4, 5\}$)

	(1)
Aspirations ratio	27.473***
	(10.528)
Step today = 4	3.893
	(6.317)
Step today = 5	3.733
	(17.586)
Assets per capita (millions)	0.253***
	(0.033)
Dependency ratio	1.627
	(6.262)
Number of children	2.564
	(5.064)
Number of Children aged 6–12	-2.555
	(6.513)
Number of Children aged 12–18	10.260**
	(4.517)
Two-parent household dummy	-16.746**
	(7.349)
log(Govt. assistance received per child)	-0.466
	(0.423)
Urban dummy	31.915***
	(5.794)
Age of household head	-0.176
	(0.321)
Years of education of household head	0.924**
	(0.428)
Head marriage status dummy	22.193
	(13.788)
Religion of head dummies	Yes
Ethnicity of head dummies	Yes
Kecamatan FE	Yes
S.E.: Clustered	EA
R^2	0.09842
Observations	3,283

Notes: Dependent variable is $100 \times \log$(Education expenditure per capita). Sample restricted to households with initial step $sw01 \in \{3, 4, 5\}$. Assets per capita measured in millions. Standard errors clustered at the EA level in parentheses. *, **, *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

A Supplementary Tables

Table A1: Distribution of household head ethnicity.

Ethnicity of head	Count	Share (%)
Javanese	2,006	43.0
Sundanese	615	13.2
Sasak	245	5.2
Minang	222	4.8
Bali	192	4.1
Bugis	188	4.0
Banjar	181	3.9
Other Southern Sumatrans	168	3.6
Batak	164	3.5
Betawi	158	3.4
Bima–Dompu	125	2.7
Maduranese	106	2.3
Makasar	74	1.6
Palembang	43	0.9
Toraja	29	0.6
Sumbawa	28	0.6
Melayu	28	0.6
Nias	19	0.4
Komering	18	0.4
Chinese	16	0.3
Others	16	0.3
Banten	15	0.3
Acehnese	6	0.1
Dayak	2	0.0
Cirebon	2	0.0
Missing / unknown	2	0.0
Manado	1	0.0
Total	4,669	100.0

Notes: Shares are expressed as percentages. Sample corresponds to the main regression sample in table 6, column (1).

Table A2: Rural households subsample.

	Main	Normalised	Sub-sample
Aspirations ratio	34.049*	31.956*	26.933
	(19.958)	(19.267)	(20.127)
(Aspirations ratio)^2	-3.477	-3.189	-2.581
	(3.185)	(3.093)	(3.151)
Step today = 2	6.598	-61.263***	7.771
	(17.534)	(17.314)	(16.999)
Step today = 3	35.954*	-73.545***	38.895**
	(18.445)	(18.273)	(17.912)
Step today = 4	44.205**	-93.828***	
	(18.657)	(18.356)	
Step today = 5	17.892	-139.239***	
	(38.942)	(35.473)	
Step today = 6	-72.558	-238.788***	
	(95.269)	(82.498)	
Assets per capita (millions)	0.185***	0.181***	0.114
	(0.068)	(0.066)	(0.079)
Dependency ratio	-5.329	-4.202	3.309
	(10.643)	(9.633)	(7.423)
Number of children	-1.125	-1.313	-6.771
	(6.885)	(6.480)	(7.068)
Number of Children aged 6–12	2.179	0.849	1.548
	(8.040)	(7.674)	(9.686)
Number of Children aged 12–18	8.784	9.151	9.484
	(5.908)	(5.713)	(6.311)
Two-parent household dummy	-23.591**	-21.803**	-15.287
	(10.804)	(10.307)	(10.114)
log(Govt. assistance received per child)	-0.065	-0.106	-0.383
	(0.596)	(0.575)	(0.686)
Age of household head	0.220	0.209	0.343
	(0.333)	(0.319)	(0.382)
Years of education of household head	1.587***	1.533***	1.580**
	(0.595)	(0.572)	(0.696)
Head marriage status dummy	28.448*	26.365*	36.888*
	(16.408)	(15.594)	(20.762)
Religion of head dummies	Yes	Yes	No
Ethnicity of head dummies	Yes	Yes	No
Kecamatan FE	Yes	Yes	Yes
S.E.: Clustered	EA	EA	EA
R^2	0.09351	0.11683	0.07668
Observations	1,969	1,969	1,564

Notes: Dependent variable is in $100 \times \log(\cdot)$. Main specification follows from eq. (5), normalised specification follows from the normalised specification and the subsample includes households on economic step today $1, 2, 3$. Standard errors in parentheses. *, **, *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

Table A3: Summary statistics for the list of controls used.

Variable	N	Mean	SD	Min	Max
Assets per capita (millions)	4651	41.1313694	67.0472219	0.0000000	985.00000
Dependency ratio	4651	0.9240161	0.5656802	0.1428571	5.00000
Head age	4651	42.8505698	9.5732731	15.0000000	87.00000
Head years of education	4651	3.0539669	6.2631102	0.0000000	16.00000
Head marriage status dummy	4651	0.9187272	0.2732831	0.0000000	1.00000
Urban dummy	4651	0.5766502	0.4941429	0.0000000	1.00000
log(Govt. assistance pe child)	4651	3.7683347	5.8242399	0.0000000	18.42068
Number of children (12 to 18)	4651	0.7047947	0.7520128	0.0000000	5.00000
Number of children (6 to 11)	4651	0.9436680	0.6374076	0.0000000	6.00000
Number of children	4651	1.8886261	0.8342543	1.0000000	10.00000
Two parent household dummy	4651	0.8426145	0.3642030	0.0000000	1.00000

Table A4: Wald test results: Joint significance linear and squared term.

Specification	Wald F	df1	df2	p-value
table 6, col (3)	5.300	2	4515	0.005
table 7, col (3)	5.478	2	4515	0.004
table 9, col (1)	5.260	2	3458	0.005
table 9, col (2)	3.135	2	1246	0.044
table 8, col(1)	2.590	2	2563	0.075
table 8, col (2)	2.714	2	2563	0.066
table 8, col (3)	3.150	2	1910	0.043
table 10	3.331	2	927	0.036

Table A5: Summary statistics (Economic step today $\{1, 2\}$).

variable	N	Mean	SD	Min	Max
Assets per capita (millions)	1329	19.2816617	30.3786480	0.0000000	368.63333
Dependency ratio	1329	0.9764932	0.6169037	0.1666667	5.00000
Aspirations ratio	1329	1.8043642	0.9561684	1.0000000	6.00000
Age of head	1329	43.2859293	10.2864598	23.0000000	87.00000
Years of education (head)	1329	1.5500376	4.6980291	0.0000000	16.00000
Head marriage status dummy	1329	0.9074492	0.2899110	0.0000000	1.00000
Urban dummy	1329	0.4981189	0.5001847	0.0000000	1.00000
log(Govt. assistance per capita)	1329	4.7193860	6.1472538	0.0000000	15.09644
Number of children (12-18)	1329	0.7426637	0.7653996	0.0000000	4.00000
Number of children (6-11)	1329	0.9699022	0.6533026	0.0000000	4.00000
Number of children	1329	1.9616253	0.8811287	1.0000000	6.00000
Future economic step	1329	2.9789315	1.1265585	1.0000000	6.00000
Economic step today	1329	1.7562077	0.4295305	1.0000000	2.00000
2 parent household dummy	1329	0.8367193	0.3697607	0.0000000	1.00000

Table A6: Summary statistics (Economic step today $\{1, 2, 3\}$).

variable	N	Mean	SD	Min	Max
Assets per capita (millions)	3556	32.6294588	55.3114466	0.0000000	917.9999967
Dependency ratio	3556	0.9318436	0.5657912	0.1428571	5.0000000
Aspirations ratio	3556	1.5254968	0.6582021	1.0000000	6.0000000
Age of head	3556	42.7871204	9.6259706	16.0000000	87.0000000
Years of education of head	3556	2.4699100	5.7540735	0.0000000	16.0000000
Head marriage status dummy	3556	0.9195726	0.2719921	0.0000000	1.0000000
Urban dummy	3556	0.5601800	0.4964350	0.0000000	1.0000000
log(Govt. assistance per capita)	3556	3.9973782	5.9187940	0.0000000	18.4206807
Number of children (12-18)	3556	0.7086614	0.7545778	0.0000000	5.0000000
Number of children (6-12)	3556	0.9507874	0.6391775	0.0000000	6.0000000
Number of children	3556	1.9063555	0.8432434	1.0000000	10.0000000
Economic step today	3556	2.5351519	0.6565918	1.0000000	3.0000000
2 parent household dummy	3556	0.8498313	0.3572870	0.0000000	1.0000000

Table A7: Summary statistics (Economic step today $\{3, 4, 5\}$).

variable	N	Mean	SD	Min	Max
Assets per capita(millions)	3283	49.4665253	74.4193101	0.0000000	985.00000
Dependency ratio	3283	0.9010882	0.5353989	0.1428571	4.00000
Aspirations ratio	3283	1.3115646	0.2491680	1.0000000	2.00000
Age of head	3283	42.6201645	9.2252493	15.0000000	83.00000
Years of education (head)	3283	3.6405726	6.6855433	0.0000000	16.00000
Head marriage status dummy	3283	0.9235455	0.2657644	0.0000000	1.00000
Urban dummy	3283	0.6073713	0.4884097	0.0000000	1.00000
log(Govt. assistance per capita)	3283	3.3864234	5.6463821	0.0000000	18.42068
Number of children (12-18)	3283	0.6908316	0.7461447	0.0000000	5.00000
Number of children (6-12)	3283	0.9311605	0.6304391	0.0000000	6.00000
Number of children	3283	1.8583613	0.8110935	1.0000000	10.00000
Future economic step	3283	4.3533354	0.8615511	3.0000000	6.00000
Economic step today	3283	3.3478526	0.5285143	3.0000000	5.00000
2 parent household dummy	3283	0.8446543	0.3622891	0.0000000	1.00000

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